

Singular Value Decomposition

Rediscovered from a question about right angles

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Abstract

The singular value decomposition is usually introduced as a factorization $A = U\Sigma V^T$, justified by applying the spectral theorem to $A^T A$. That is correct, and it explains nothing: the geometry arrives already compressed into an algebraic identity, and the reader is asked to accept a powerful theorem at the outset in order to reach a result that is, in the end, about ellipses.

This note reverses the order. It starts with a question one can ask about any linear map — *which right angles does it preserve?* — and answers it first in the plane, where the whole situation is visible: a perpendicular frame is rotated, its images skew, the skew reverses sign, and somewhere in between the right angle survives. That crossing turns out to be the same point at which the map stretches hardest, an equivalence one can watch happen and then prove both ways.

Only one half of that equivalence survives the trip to $\mathbb{R}^n$, and it is the half that matters: at a maximizer, the image cannot tilt. Iterating this yields the full frame in any dimension, with the singular values falling out in descending order because each step maximizes over what the last one left. The construction then turns out to have proved something it was not aiming at — its output is an orthonormal eigenbasis of $A^T A$, which is to say the spectral theorem, obtained rather than assumed. The classical statement appears at the end as a name for what the geometry already built, not as a hypothesis at the start.

The decomposition's existence, uniqueness, parameter count, degenerate cases, and relation to the eigendecomposition all follow from the same source. Worked examples are small enough to verify by hand; every numerical and symbolic claim has been checked.

1 The question

Fix a matrix $A \in \mathbb{R}^{m \times n}$ and think of it as a map $\mathbb{R}^n \rightarrow \mathbb{R}^m$. Linear maps do not, in general, preserve angles. Take

$$A = \begin{pmatrix} 2 & 1 \\ -0.5 & 1.5 \end{pmatrix},$$

which will serve as the running example. The standard basis vectors e_1, e_2 start out perpendicular; their images $Ae_1 = (2, -0.5)$ and $Ae_2 = (1, 1.5)$ have inner product 1.25 and sit 70.35° apart. The right angle is destroyed.

Try a different perpendicular pair and you get a different amount of damage — sometimes more, sometimes less. This suggests a question:

Is there a perpendicular pair whose images are still perpendicular?

The answer is yes, for every matrix without exception, and the pair is essentially unique. Finding it *is* the singular value decomposition.

The route there runs through three stages, each earning the next. §§2.1–2.2 settle the plane, where the answer can be watched rather than computed. §2.3 extracts from that picture the one statement that survives into $\mathbb{R}^n$, and iterates it into a construction valid in any dimension. §2.4 observes that the construction has already proved the spectral theorem, and only then states it — as a name for what has been built, rather than as an assumption. Everything afterwards follows from the frame: what it is called (§3), what it looks like (§4), and what can be deduced from having found it (§§6–9).

2 Why such a pair must exist

Take an orthonormal frame $\mathbf{v}_1(\theta), \mathbf{v}_2(\theta)$ and rotate it rigidly as θ varies. Define

$$f(\theta) = \langle A\mathbf{v}_1(\theta), A\mathbf{v}_2(\theta) \rangle,$$

the inner product of the two *images*. This is the quantity we want to vanish (Figure 1): $f > 0$ means the images have closed to an acute angle, $f < 0$ means they have opened past a right angle, and $f = 0$ means the right angle survived. Note that f measures the *damage*, not the angle itself; the input frame stays perpendicular throughout, by construction.

2.1 The sign flips every quarter turn

Rotate the frame by exactly 90° . What happens to its two arms?

- $\mathbf{v}_1$ rotated by 90° becomes $\mathbf{v}_2$. That is what perpendicular means.
- $\mathbf{v}_2$ rotated by 90° becomes $-\mathbf{v}_1$. Another quarter turn past $\mathbf{v}_2$ lands *opposite* $\mathbf{v}_1$, not on it.

So $\mathbf{v}_1(\theta + 90^\circ) = \mathbf{v}_2(\theta)$ and $\mathbf{v}_2(\theta + 90^\circ) = -\mathbf{v}_1(\theta)$. Apply A , use linearity to pull the sign out, and use the symmetry of the inner product:

$$f(\theta + 90^\circ) = \langle A\mathbf{v}_2, A(-\mathbf{v}_1) \rangle = -\langle A\mathbf{v}_2, A\mathbf{v}_1 \rangle = -\langle A\mathbf{v}_1, A\mathbf{v}_2 \rangle = -f(\theta).$$

This is worth pausing on, because the swap alone would achieve nothing — the inner product is symmetric, so exchanging its two arguments leaves it unchanged. *The minus sign is doing all the work.* And that minus sign is unavoidable: you cannot rotate an orthonormal pair by a quarter turn and recover the same pair, because one arm always comes back reversed.

Claim. *Every matrix preserves at least one right angle.*

Proof. f is continuous, and $f(\theta_0 + 90^\circ) = -f(\theta_0)$ for any θ_0 . If $f(\theta_0) = 0$ we are done. Otherwise $f(\theta_0)$ and $f(\theta_0 + 90^\circ)$ have opposite signs, so by the intermediate value theorem f vanishes somewhere strictly between. $\square$

The argument used nothing about A — not invertibility, not squareness, not any structure. In $\mathbb{R}^n$ the same conclusion holds via the spectral theorem (§2.4); the planar version is just the case you can see.

2.2 How many such angles are there?

Continuity gives existence but is silent on count. To settle that, compute f in closed form. Writing $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{v}_1 = (\cos \theta, \sin \theta)$, $\mathbf{v}_2 = (-\sin \theta, \cos \theta)$, expanding and collecting terms gives

$$f(\theta) = (ab + cd) \cos 2\theta - \frac{1}{2} (a^2 - b^2 + c^2 - d^2) \sin 2\theta$$

Two features of this expression matter, and the second is the important one.

First, everything is in 2θ , not θ . That is the algebraic shadow of the quarter-turn symmetry: advancing θ by 90° advances 2θ by 180° , which negates both $\cos$ and $\sin$. The sign flip proved above is visible directly in the formula.

Second, and crucially, there is no constant term. f is a pure sinusoid oscillating about zero — not a sinusoid plus an offset. This is why the sign flip is not merely a curiosity but a guarantee: a sinusoid with a nonzero mean might never reach zero, but this one must.

Write it in amplitude–phase form as

$$f(\theta) = R \cos(2\theta - \varphi), \quad R = \sqrt{A_1^2 + A_2^2}, \quad \varphi = \text{atan2}(A_2, A_1),$$

where $A_1 = ab + cd$ and $A_2 = -\frac{1}{2}(a^2 - b^2 + c^2 - d^2)$. Suppose $R \neq 0$. As θ runs over a half turn $[0^\circ, 180^\circ)$, the argument 2θ runs over a *full* turn, so the cosine vanishes exactly twice, at

$$\theta = \frac{\varphi \pm 90^\circ}{2}.$$

The two roots are 90° apart (Figure 1). But a frame and the same frame rotated by 90° consist of *the same pair of lines* — as shown above, the quarter turn merely swaps the arms and reverses one. So the two roots describe one geometric object.

Claim. *Unless $R = 0$, the frame is unique: exactly one perpendicular pair of directions has perpendicular images, up to relabelling the arms and flipping their signs.*

The degenerate case. $R = 0$ means $A_1 = A_2 = 0$, i.e. $f \equiv 0$ and *every* frame works. The two coefficients are readable off $A^\top A$:

$$A^\top A = \begin{pmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{pmatrix},$$

so A_1 is its off-diagonal entry and $-2A_2$ the difference of its diagonal entries. Both vanish precisely when $A^\top A = \sigma^2 I$ — when A is a scalar multiple of an orthogonal matrix. Then A maps the sphere to a sphere, there is no direction of maximum stretch, and no frame is distinguished. The generic case and the degenerate case both fall out of the one formula.

2.3 From two dimensions to n

The rotating frame is a planar device. In $\mathbb{R}^n$ there is no single angle to turn: an orthonormal frame has $n(n-1)/2$ parameters, the images must satisfy $n(n-1)/2$ orthogonality conditions, and the intermediate value theorem — one equation, one unknown — has nothing to say about a coupled system that size. Fixing up one pair disturbs another.

This section builds the bridge. The plan is to establish, *in the plane where both sides can be seen*, that the surviving frame and the direction of maximum stretch are the same object; then to notice that one half of that equivalence never used the dimension; then to recurse on it.

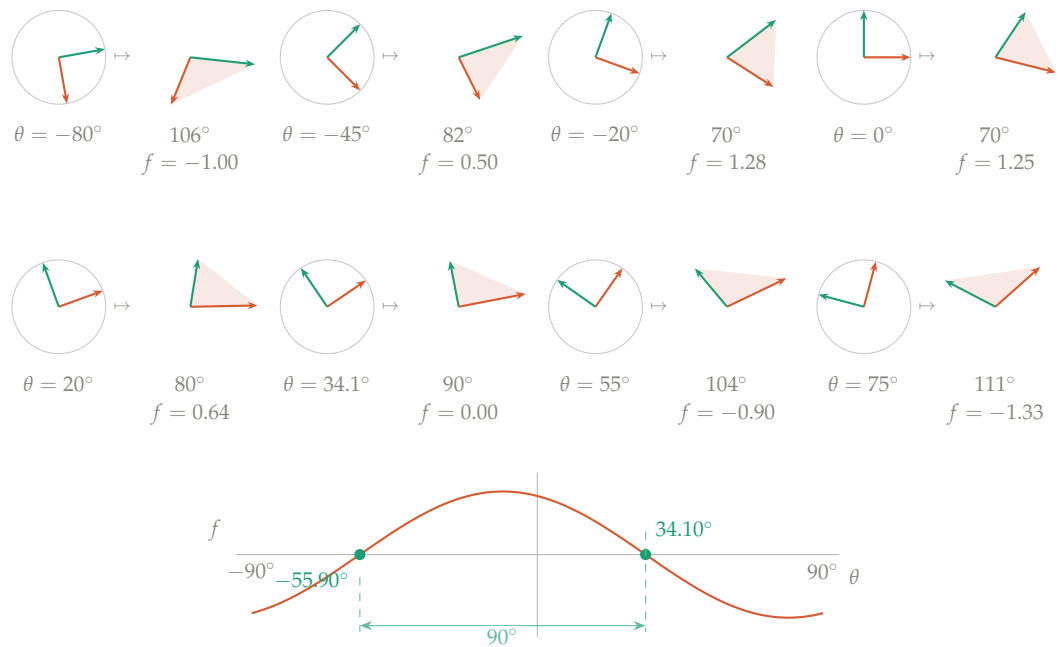


Figure 1: **Sweeping the frame.** Each panel shows an orthonormal input frame (left, always perpendicular) and its image under $A = \begin{pmatrix} 2 & 1 \\ -0.5 & 1.5 \end{pmatrix}$ (right, generally not). The number below each image is the angle between $A\mathbf{v}_1$ and $A\mathbf{v}_2$; the shaded wedge makes it visible. As θ increases the wedge narrows to 70° , then widens back through 90° — the highlighted panel, where the right angle survives — and on past 110° . The plot below tracks $f(\theta)$ over the full half-turn: a pure sinusoid in 2θ with no vertical offset, crossing zero exactly twice, at 34.10° and -55.90° . Those two crossings are 90° apart and describe the same pair of lines.

A false start, worth naming. It is tempting to hope that orthogonality in the domain forces orthogonality in the image, or the reverse. Neither holds. The shear sends the perpendicular pair e_1, e_2 to $(1, 0)$ and $(1, 1)$, which are 45° apart; and pulling a perpendicular pair back through A^{-1} generically yields a skew pair. Perpendicularity is not preserved in either direction — which is exactly why the question “is there *some* frame it is preserved for?” has content. The domain-side right angle in $f(\theta)$ is a constraint we impose, not a fact we derive.

In the plane: the crossing is the peak

Recall $f(\theta) = \langle A\mathbf{v}_1(\theta), A\mathbf{v}_2(\theta) \rangle$ and let

$$g(\theta) = \|A\mathbf{v}_1(\theta)\|^2$$

be the squared stretch of the first arm. Differentiating g and comparing with f gives an identity worth stating on its own line:

$$\boxed{\frac{dg}{d\theta} = 2f(\theta)}$$

The damage function is *half the derivative of the stretch*. So the zero crossing of Figure 1 is not merely correlated with the ellipse’s axis — it *is* the critical point of g . Watching the right angle survive and watching the stretch peak are the same observation, made with different instruments.

That is the equivalence, and both directions are worth having explicitly.

Maximality $\Rightarrow$ orthogonal images. Let $\mathbf{v}_1$ maximize $\|Av\|$ over unit vectors, and let w be any unit vector with $w \perp \mathbf{v}_1$. Slide from one to the other along the unit sphere:

$$v(t) = \cos t \mathbf{v}_1 + \sin t w, \quad \|v(t)\| = 1.$$

Expanding $\|Av(t)\|^2 = \langle Av(t), Av(t) \rangle$ and differentiating at $t = 0$,

$$\left. \frac{d}{dt} \right|_{t=0} \|Av(t)\|^2 = 2\langle A\mathbf{v}_1, Aw \rangle.$$

But $t = 0$ is a maximum of a differentiable function, so the derivative vanishes there. Hence

$$\langle A\mathbf{v}_1, Aw \rangle = 0 \quad \text{for every } w \perp \mathbf{v}_1.$$

The images are orthogonal *because $\mathbf{v}_1$ is a peak*: if $A\mathbf{v}_1$ had any component tilting toward Aw , leaning $\mathbf{v}_1$ a little toward w would buy more stretch, contradicting maximality.

Orthogonal images $\Rightarrow$ maximality. Conversely, suppose $\langle A\mathbf{v}_1, A\mathbf{v}_2 \rangle = 0$ and label so that $\sigma_1 \geq \sigma_2$. Any unit vector is $v = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ with $c_1^2 + c_2^2 = 1$, and the orthogonality of the images collapses the cross terms:

$$\|Av\|^2 = \|c_1A\mathbf{v}_1 + c_2A\mathbf{v}_2\|^2 = c_1^2\sigma_1^2 + \underbrace{2c_1c_2\langle A\mathbf{v}_1, A\mathbf{v}_2 \rangle}_{=0} + c_2^2\sigma_2^2 = c_1^2\sigma_1^2 + c_2^2\sigma_2^2.$$

This is a weighted average of σ_1^2 and σ_2^2 with weights summing to one, so it is largest when all the weight sits on the larger value: $c_1 = \pm 1$, i.e. $v = \pm \mathbf{v}_1$. The surviving frame’s first arm is the maximizer.

The hinge

Now look back at the first of those two arguments and ask what it used. A unit vector $\mathbf{v}_1$; a perpendicular unit vector w ; the curve $\cos t \mathbf{v}_1 + \sin t w$; one derivative. *Nothing in it mentions the dimension.* In $\mathbb{R}^2$ the only choice of w is $\pm \mathbf{v}_2$, so the conclusion looks like a statement about a single pair. In $\mathbb{R}^n$ the same curve lies on the unit sphere S^{n-1} , the same derivative computation goes through unchanged, and the conclusion becomes considerably stronger:

$$\langle A\mathbf{v}_1, Aw \rangle = 0 \quad \text{for every } w \perp \mathbf{v}_1,$$

which is $n - 1$ orthogonality relations at once, not one. The maximizer's image is orthogonal to the image of the entire hyperplane $\mathbf{v}_1^\perp$.

The other direction does not need to make the trip. It was there to show that the two planar descriptions coincide; having established that, we keep only the half that travels.

In $\mathbb{R}^n$: recursion

The construction now runs by induction, and each step is the hinge applied once.

1. The function $v \mapsto \|Av\|$ is continuous and S^{n-1} is compact, so a maximizer exists. Call it $\mathbf{v}_1$ and set $\sigma_1 = \|A\mathbf{v}_1\|$.
2. By the hinge, $\langle A\mathbf{v}_1, Aw \rangle = 0$ for every $w \perp \mathbf{v}_1$. So *whatever* we choose next from $\mathbf{v}_1^\perp$, its image is already orthogonal to $A\mathbf{v}_1$ — for free, with no further work.
3. Restrict A to the hyperplane $\mathbf{v}_1^\perp \cong \mathbb{R}^{n-1}$ and repeat: maximize $\|Av\|$ over the unit sphere of $\mathbf{v}_1^\perp$ to get $\mathbf{v}_2$, with $\sigma_2 = \|A\mathbf{v}_2\| \leq \sigma_1$. Applying the hinge *within* that hyperplane gives $\langle A\mathbf{v}_2, Aw \rangle = 0$ for all $w \in \mathbf{v}_1^\perp$ with $w \perp \mathbf{v}_2$.
4. Continue. After n steps the frame $\mathbf{v}_1, \dots, \mathbf{v}_n$ is orthonormal by construction, and its images are pairwise orthogonal: for $i < j$, step i 's hinge applies because $\mathbf{v}_j$ lies in $\mathbf{v}_i^\perp$.

Two features of this deserve comment. First, the σ_i come out *sorted*, $\sigma_1 \geq \sigma_2 \geq \dots$, automatically: each step maximizes over a subset of the previous step's domain, so it cannot do better. The descending convention is not a convention here — it is what the construction produces. Second, step 2 is doing all the work, and it is doing it *before* the next vector is chosen: maximality of $\mathbf{v}_1$ pre-emptively guarantees orthogonality against every future choice. That is what makes the recursion compose, and it is precisely what the planar picture could not show, having no room to recurse into.

What the construction has already proved

Something has quietly happened. Return to the hinge, the statement that at the maximizer

$$\langle A\mathbf{v}_1, Aw \rangle = 0 \quad \text{for every } w \perp \mathbf{v}_1,$$

and move A across the inner product onto the other argument:

$$\langle A\mathbf{v}_1, Aw \rangle = (A\mathbf{v}_1)^\top (Aw) = \mathbf{v}_1^\top A^\top A w = \langle A^\top A \mathbf{v}_1, w \rangle.$$

So the hinge says: *the vector $A^\top A \mathbf{v}_1$ is orthogonal to every w in the hyperplane $\mathbf{v}_1^\perp$* . But the only vectors orthogonal to all of $\mathbf{v}_1^\perp$ are the multiples of $\mathbf{v}_1$ itself. Therefore

$$A^\top A \mathbf{v}_1 = \lambda_1 \mathbf{v}_1$$

for some scalar λ_1 — and pairing both sides with $\mathbf{v}_1$ identifies it,

$$\lambda_1 = \langle A^\top A \mathbf{v}_1, \mathbf{v}_1 \rangle = \langle A \mathbf{v}_1, A \mathbf{v}_1 \rangle = \|A \mathbf{v}_1\|^2 = \sigma_1^2.$$

The maximizer is an eigenvector of $A^\top A$, with eigenvalue the squared stretch. This was not assumed. It fell out of maximality alone, by one application of the adjoint move.

And the same holds at every step of the recursion, since each $\mathbf{v}_i$ maximizes over a subspace on which the identical argument runs. So the construction of the previous page did not merely find a convenient frame — it produced an orthonormal basis $\mathbf{v}_1, \dots, \mathbf{v}_n$ of eigenvectors of $A^\top A$, with nonnegative eigenvalues σ_i^2 , sorted. That is a complete diagonalization:

$$A^\top A = V \Sigma^2 V^\top.$$

Which is to say: *the variational route has already proved the spectral theorem for $A^\top A$* . The next section does not introduce new machinery. It names this result in the standard language, states it for symmetric matrices in general rather than for Gram matrices in particular, and notes that once one has it, the entire construction collapses to a single sentence.

One caveat. The route above is specific to matrices of the form $A^\top A$, because it maximizes $\|Ax\|^2$, a quantity that is automatically nonnegative. A general symmetric S can have negative eigenvalues — for instance $\begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix}$ has eigenvalues -3.19 and 2.19 — so no maximization of a squared length will ever produce them. The fix is to run the same argument on the *Rayleigh quotient*

$$\rho(x) = \frac{x^\top S x}{x^\top x},$$

whose stationary points are exactly the eigenvectors of S and whose stationary values are the eigenvalues. The proof is the same derivative computation; only the functional changes. So the spectral theorem in full generality is this section's argument with $\|Ax\|^2$ replaced by $\rho(x)$, and $\rho(x) = \|Ax\|^2$ on the unit sphere when $S = A^\top A$ recovers what we did.

2.4 The same fact, algebraically

The previous section arrived at $A^\top A \mathbf{v}_i = \sigma_i^2 \mathbf{v}_i$ with the $\mathbf{v}_i$ orthonormal. That statement has a name, and stating it for a general symmetric matrix rather than for $A^\top A$ costs nothing — the hypothesis that actually does the work is symmetry, not the particular form of the matrix.

Spectral theorem. *Let S be a real symmetric $n \times n$ matrix, $S = S^\top$. Then S has n real eigenvalues $\lambda_1, \dots, \lambda_n$ (with multiplicity), and $\mathbb{R}^n$ admits an orthonormal basis $\mathbf{w}_1, \dots, \mathbf{w}_n$ of eigenvectors, $S \mathbf{w}_i = \lambda_i \mathbf{w}_i$. Equivalently $S = W \Lambda W^\top$ with W orthogonal and Λ diagonal.*

Two clauses matter here and neither is free. *Real* eigenvalues: a general real matrix can have complex ones, as a rotation does. *Orthonormal* eigenbasis: a general matrix's eigenvectors need not be perpendicular, and a defective matrix does not have a full set of them at all. Symmetry buys both.

Taking the theorem as given — it is standard, and §2.3 sketched its proof — the whole of §2 and §2.3 collapses to three lines, which is the point of this section. The rotating frame and the greedy maximization were two ways of finding by hand what the following computation supplies at once, in any dimension.

Where the symmetry enters

The orthogonality half is short enough to see in full. Let $\mathbf{w}_i, \mathbf{w}_j$ be eigenvectors with $\lambda_i \neq \lambda_j$. Start from the number $\langle S\mathbf{w}_i, \mathbf{w}_j \rangle$ and evaluate it two ways.

Pushing S onto the left argument:

$$\langle S\mathbf{w}_i, \mathbf{w}_j \rangle = \langle \lambda_i \mathbf{w}_i, \mathbf{w}_j \rangle = \lambda_i \langle \mathbf{w}_i, \mathbf{w}_j \rangle \quad (\text{eigenvector equation, then linearity}).$$

Moving S across the inner product to the right argument instead:

$$\langle S\mathbf{w}_i, \mathbf{w}_j \rangle = (S\mathbf{w}_i)^\top \mathbf{w}_j = \mathbf{w}_i^\top S^\top \mathbf{w}_j = \underbrace{\mathbf{w}_i^\top S \mathbf{w}_j}_{\text{here } S^\top = S} = \langle \mathbf{w}_i, S\mathbf{w}_j \rangle = \lambda_j \langle \mathbf{w}_i, \mathbf{w}_j \rangle.$$

The symmetry hypothesis is used exactly once, at the brace: without $S^\top = S$ the chain breaks and nothing follows. Setting the two evaluations equal,

$$\lambda_i \langle \mathbf{w}_i, \mathbf{w}_j \rangle = \lambda_j \langle \mathbf{w}_i, \mathbf{w}_j \rangle \implies (\lambda_i - \lambda_j) \langle \mathbf{w}_i, \mathbf{w}_j \rangle = 0.$$

Since $\lambda_i \neq \lambda_j$ by assumption, the first factor is nonzero, so $\langle \mathbf{w}_i, \mathbf{w}_j \rangle = 0$: the eigenvectors are perpendicular. (If $\lambda_i = \lambda_j$ the argument says nothing, and correctly so — within a repeated eigenvalue's eigenspace any basis is an eigenbasis, so one simply chooses an orthonormal one. This is the same freedom catalogued in §7 for repeated singular values.)

Applying it to $A^\top A$

First, $A^\top A$ qualifies. It is symmetric because transposing reverses the order of a product and undoes itself:

$$(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A.$$

The matrix $A^\top A$ is called the **Gram matrix** of A : its (i, j) entry is the inner product of column i of A with column j , so its diagonal holds the squared column lengths and its off-diagonal entries measure how far from perpendicular the columns are. (Throughout, $(M)_{ij}$ denotes the entry of M in row i , column j .)

So the spectral theorem applies and supplies an orthonormal eigenbasis $\mathbf{v}_1, \dots, \mathbf{v}_n$ with $A^\top A \mathbf{v}_i = \lambda_i \mathbf{v}_i$ — for free, in any dimension, with no rotating frame required. Now take any two

of them, $i \neq j$, and compute the inner product of their images one step at a time:

$$\begin{aligned}
 \langle A\mathbf{v}_i, A\mathbf{v}_j \rangle &= (A\mathbf{v}_i)^\top (A\mathbf{v}_j) && \text{definition of the inner product} \\
 &= \mathbf{v}_i^\top A^\top A \mathbf{v}_j && \text{transpose of a product} \\
 &= \mathbf{v}_i^\top (A^\top A \mathbf{v}_j) && \text{regrouping; matrix product is associative} \\
 &= \mathbf{v}_i^\top (\lambda_j \mathbf{v}_j) && \mathbf{v}_j \text{ is an eigenvector of } A^\top A \\
 &= \lambda_j \mathbf{v}_i^\top \mathbf{v}_j && \text{scalars pull out} \\
 &= \lambda_j \langle \mathbf{v}_i, \mathbf{v}_j \rangle && \text{definition again} \\
 &= 0 && \text{the } \mathbf{v}_i \text{ are orthonormal, } i \neq j.
 \end{aligned}$$

The images are orthogonal automatically. *The eigenvectors of $A^\top A$ are the frame the whole note is about*, and the rotating-frame picture of §2 is what this computation looks like in the plane.

Notice which step did the work. The eigenvector equation converted the *operator* $A^\top A$ into the *scalar* λ_j , after which orthogonality of the $\mathbf{v}_i$ finished the job. Feed in a non-eigenvector and line four fails: $A^\top A \mathbf{v}_j$ is some other vector, not a multiple of $\mathbf{v}_j$, and the inner product has no reason to vanish.

Why the singular values are real

$A^\top A$ is not merely symmetric but positive semidefinite. For any vector x ,

$$x^\top (A^\top A) x = \underbrace{(x^\top A^\top)}_{=(Ax)^\top} (Ax) = (Ax)^\top (Ax) = \|Ax\|^2 \geq 0,$$

the last step because a squared length is never negative. Now take $x = \mathbf{v}_i$, an eigenvector:

$$\mathbf{v}_i^\top (A^\top A) \mathbf{v}_i = \mathbf{v}_i^\top (\lambda_i \mathbf{v}_i) = \lambda_i \underbrace{\mathbf{v}_i^\top \mathbf{v}_i}_{=\|\mathbf{v}_i\|^2=1} = \lambda_i,$$

and the same quantity equals $\|A\mathbf{v}_i\|^2$ by the display above. Hence

$$\lambda_i = \|A\mathbf{v}_i\|^2 \geq 0,$$

so $\sigma_i := \sqrt{\lambda_i}$ is a well-defined nonnegative real number, and it is exactly the stretch factor $\|A\mathbf{v}_i\|$ we wanted.

This step deserves its own paragraph because it is easy to skip past. Symmetry alone gives *real* eigenvalues, but real numbers can be negative, and $\sqrt{\lambda_i}$ would then be imaginary — Σ would not exist. What rules that out is positive semidefiniteness, which comes from $A^\top A$ being a Gram matrix rather than from its symmetry. Both properties are needed, and they are different properties.

The same statement in the plane

The two arguments are not merely analogous; they are the same equation. To see it, take any symmetric 2×2 matrix $G = \begin{pmatrix} p & q \\ q & r \end{pmatrix}$ and ask what its entries become in a frame rotated by θ . Conjugating by the rotation $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ and reading off the off-diagonal entry gives

$$\left(R_\theta^\top G R_\theta \right)_{12} = q (\cos^2 \theta - \sin^2 \theta) + (r - p) \sin \theta \cos \theta = q \cos 2\theta + \frac{r - p}{2} \sin 2\theta.$$

Now substitute the Gram matrix, $G = A^T A$, whose entries are

$$p = a^2 + c^2, \quad q = ab + cd, \quad r = b^2 + d^2.$$

Then $q = ab + cd$ and $\frac{r-p}{2} = -\frac{1}{2}(a^2 - b^2 + c^2 - d^2)$, so

$$\left(R_\theta^T A^T A R_\theta\right)_{12} = (ab + cd) \cos 2\theta - \frac{1}{2}(a^2 - b^2 + c^2 - d^2) \sin 2\theta = f(\theta),$$

matching the closed form of §2.2 term for term. The off-diagonal entry of the Gram matrix in the rotated frame is the damage function.

So diagonalizing $A^T A$ and solving $f(\theta) = 0$ are not two routes to the same answer — they are one problem stated twice. Both ask for the rotation that annihilates the off-diagonal entry. And the diagonal entries left behind,

$$\left(R_\theta^T A^T A R_\theta\right)_{11} = \|A\mathbf{v}_1\|^2 = \sigma_1^2, \quad \left(R_\theta^T A^T A R_\theta\right)_{22} = \|A\mathbf{v}_2\|^2 = \sigma_2^2,$$

are the squared singular values. For the running example this is checked numerically in §5.

Three routes, one frame

	Shows	Costs
Rotating frame (§2)	why a crossing is forced, and where it sits: the ful- crum where the distortion cancels	planar only; no route to $\mathbb{R}^n$
Variational (§2.3)	constructive; explains why the σ_i come out sorted; proves the spectral theorem rather than assuming it	no picture above $\mathbb{R}^2$; sta- tionarity replaces balance
Spectral (§2.4)	fastest; every dimension at once	most opaque; the geometry is invisible in the statement

These are not three competing theorems. They are one theorem approached from three distances. The variational route is the spectral theorem's proof with the geometry left in; the spectral statement is that proof with the geometry compressed out; and the rotating frame is what both look like in the one dimension where you can watch them happen. The planar picture is worth keeping even though it does not scale, because it shows what the machinery above it is *for*: the greedy maximization and the eigenvectors are both, in the end, hunting the right angle that survives.

3 Naming the pieces

Write $\mathbf{v}_1, \dots, \mathbf{v}_n$ for that special input frame — the **right singular vectors**. Their images are orthogonal but not unit length, so factor out the lengths:

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i, \quad \sigma_i = \|A\mathbf{v}_i\| \geq 0, \quad \|\mathbf{u}_i\| = 1.$$

The σ_i are the **singular values** and the $\mathbf{u}_i$ are the **left singular vectors**. Collecting these n equations into matrix form gives $AV = U\Sigma$, i.e.

$$A = U\Sigma V^T$$

with U, V orthogonal and Σ diagonal. The names *left* and *right* are positional — U sits on the left of the product, V on the right — and carry no deeper meaning.

Note the asymmetry in how the two frames arise. *V is chosen; U is a consequence.* We select the input frame by a property it has (its images are orthogonal), and the output frame is then whatever we happen to land on.

Four ways to say the same thing

The right singular vectors admit several characterizations, all equivalent:

Geometric. The orthonormal input frame whose images stay orthogonal. Every other orthonormal frame comes out skewed.

Metric. The frame that lands on the principal axes of the image ellipsoid (§4), with σ_i the semi-axis lengths.

Variational. The frame obtained by greedy maximization: $\mathbf{v}_1 = \arg \max_{\|v\|=1} \|Av\|$, then $\mathbf{v}_2$ the same among unit vectors orthogonal to $\mathbf{v}_1$, and so on.

Algebraic. The eigenvectors of $A^T A$.

The first is the one to keep in your head; the last is the one you compute with. Symmetrically, the left singular vectors are the eigenvectors of AA^T . Since $A^T A$ and AA^T share the same nonzero eigenvalues σ_i^2 , one Σ serves both frames.

4 The ellipsoid

Let $S^{n-1} = \{v \in \mathbb{R}^n : \|v\| = 1\}$ be the unit sphere in the input space. Its image $A(S^{n-1})$ is an ellipsoid. To see this, write v in the V -basis as $v = \sum c_i \mathbf{v}_i$ with $\sum c_i^2 = 1$. Then $Av = \sum c_i \sigma_i \mathbf{u}_i$, so the image point has coordinate $y_i = \sigma_i c_i$ along $\mathbf{u}_i$, and the constraint becomes

$$\sum_i \frac{y_i^2}{\sigma_i^2} = 1,$$

the standard ellipsoid equation in u -coordinates. The $\mathbf{u}_i$ are the axis *directions*; the σ_i are the axis *lengths*. Neither alone describes the ellipsoid — you need both.

This gives the reading of the factorization as a sequence of three motions. Right to left in $U\Sigma V^T$:

V^T	rotate the input so that $\mathbf{v}_i$ lands on the i th coordinate axis
Σ	stretch axis i by σ_i
U	rotate so that the coordinate axes land on the $\mathbf{u}_i$

Every linear map is a rotation, then an axis-aligned scaling, then another rotation. A shear does not look like that — it looks like sliding — but it is one, and the SVD is the change of viewpoint that reveals it.

Degenerate cases

The word “ellipsoid” must be read loosely. If some $\sigma_i = 0$ the ellipsoid is flattened in that direction. If $m > n$ it is a lower-dimensional ellipsoid embedded in a larger space. If $m < n$ the sphere maps onto the *solid* ellipsoid, not just its surface, because vectors with a kernel component land strictly inside.

The statement that covers every case at once: for A of rank r ,

A maps the unit sphere of the *row space* bijectively onto an $(r-1)$ -dimensional ellipsoid surface in the *column space*, with semi-axes $\sigma_1, \dots, \sigma_r$ along $\mathbf{u}_1, \dots, \mathbf{u}_r$. The kernel is crushed to zero; the cokernel is never reached.

This is the four-fundamental-subspaces picture with metric information attached. The SVD says A is an isomorphism from row space to column space, and diagonalizes it. Explicitly, with $r = \text{rank } A$:

$$\underbrace{\mathbf{v}_1 \dots \mathbf{v}_r}_{\text{row space}} \quad \underbrace{\mathbf{v}_{r+1} \dots \mathbf{v}_n}_{\text{ker } A} \quad \underbrace{\mathbf{u}_1 \dots \mathbf{u}_r}_{\text{column space}}, \quad \underbrace{\mathbf{u}_{r+1} \dots \mathbf{u}_m}_{\text{coker } A}.$$

5 A worked example

Take

$$A = \begin{pmatrix} 2 & 1 \\ -0.5 & 1.5 \end{pmatrix},$$

a matrix with no special structure — not symmetric, not a multiple of a rotation. Its Gram matrix is

$$A^T A = \begin{pmatrix} 4.25 & 1.25 \\ 1.25 & 3.25 \end{pmatrix},$$

whose off-diagonal entry is nonzero, so the standard basis is *not* the frame we want. Feeding $a = 2, b = 1, c = -0.5, d = 1.5$ into the closed form of §2.2,

$$f(\theta) = 1.25 \cos 2\theta - 0.5 \sin 2\theta,$$

with amplitude $R = 1.3463$ and phase $\varphi = -21.80^\circ$. The roots $\theta = (\varphi \pm 90^\circ)/2$ land at 34.10° and -55.90° — the crossings marked in Figure 1. Taking the first,

$$\mathbf{v}_1 = (0.8281, 0.5606), \quad \mathbf{v}_2 = (-0.5606, 0.8281),$$

and the images are

$$A\mathbf{v}_1 = (2.2168, 0.4269), \quad A\mathbf{v}_2 = (-0.2932, 1.5224).$$

Their inner product is $(2.2168)(-0.2932) + (0.4269)(1.5224) = 0$, as promised. Their lengths are the singular values,

$$\sigma_1 = 2.2575, \quad \sigma_2 = 1.5504,$$

and normalizing gives $\mathbf{u}_1 = (0.9820, 0.1891)$ and $\mathbf{u}_2 = (-0.1891, 0.9820)$. As a check, $\sigma_1 \sigma_2 = 3.5 = \det A$.

That gap of 23.20° between the frames is the rotation UV^T . Since $\det(UV^T) = +1$ it is a genuine rotation, not a reflection. For a generic matrix it is nonzero and bears no relation to anything else about A : knowing V tells you nothing about U without passing through A .

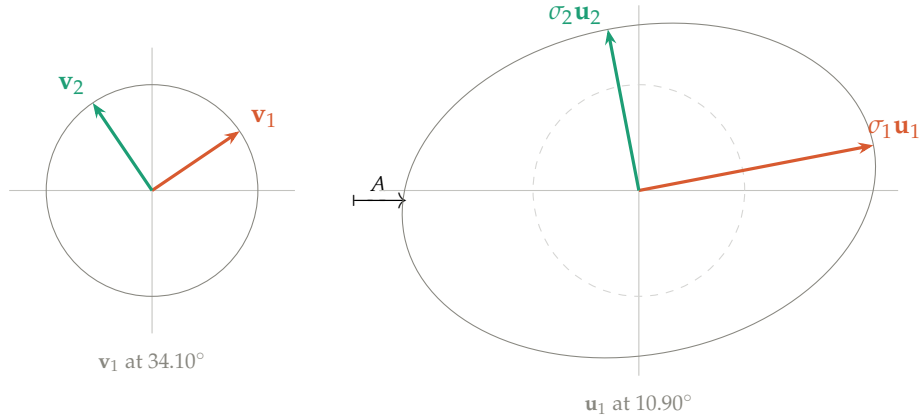


Figure 2: The unit circle and its image. The frame $\mathbf{v}_1, \mathbf{v}_2$ at 34.10° is the one whose images stay perpendicular; those images land on the ellipse's axes, with lengths $\sigma_1 = 2.2575$ and $\sigma_2 = 1.5504$. The input frame sits at 34.10° and the output frame at 10.90° : a gap of 23.20° . The two panels are different copies of $\mathbb{R}^2$; the only link between them is the pairing $\mathbf{v}_i \mapsto \sigma_i \mathbf{u}_i$.

5.1 An aside: the shear

The virtue of an arbitrary matrix is that it has no structure to mislead. The vice is that nobody has any prior expectation about it, so the SVD's claim — *this is a rotation, a stretch, and a rotation* — is unsurprising.

For that, the shear is better:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

It slides every point rightward by an amount equal to its height. Sliding does not look like rotate–stretch–rotate. But

$$A^T A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \lambda = \frac{3 \pm \sqrt{5}}{2}, \quad \sigma_1 = 1.6180, \quad \sigma_2 = 0.6180,$$

the golden ratio and its reciprocal, with $\sigma_1 \sigma_2 = 1 = \det A$. The frames sit at $\mathbf{v}_1$: 58.28° and $\mathbf{u}_1$: 31.72° , differing by a rotation of 26.57° . So the shear *is* a rotate–stretch–rotate, and the SVD is the change of viewpoint that reveals it.

Why 58.28° ? The shear pushes points rightward in proportion to height. A frame sitting in the first quadrant has both arms up high, so both get pushed right and they rotate toward each other: the angle closes, $f > 0$. A frame straddling the vertical has one arm high-right and one high-left; pushing both rightward separates them and the angle opens, $f < 0$. At 58.28° the two effects balance. The right angle sits at the fulcrum and comes through intact.

One caution about the shear, which is why it is an aside rather than the main example. Its $\mathbf{u}_1 = (0.8507, 0.5257)$ is exactly $\mathbf{v}_1 = (0.5257, 0.8507)$ with the coordinates swapped, and the two angles sum to 90° . That is an artifact: the shear is *persymmetric* (symmetric about its anti-diagonal), which forces the relation. It is easy to mistake this tidiness for alignment, or for a general fact. It is neither. The generic matrix above has no such pattern, and it is the honest picture.

	σ_1	σ_2	$\mathbf{v}_1$ angle	$\mathbf{u}_1$ angle	gap $UV^\top$
Generic $\begin{pmatrix} 2 & 1 \\ -0.5 & 1.5 \end{pmatrix}$	2.2575	1.5504	34.10°	10.90°	23.20°
Shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	1.6180	0.6180	58.28°	31.72°	26.57°

Table 1: Neither matrix has aligned frames; both rotate by roughly 25° between input and output. Alignment $U = V$ would require A symmetric positive definite (§8), and neither matrix is symmetric.

6 Can you choose the input frame?

A natural question: given A , could you pick *any* orthonormal basis V of the input space and find a matching orthonormal U and some diagonal Σ' with $A = U\Sigma'V^\top$? Perhaps the freedom to choose Σ' buys enough slack.

No. Requiring $AV = U\Sigma'$ says precisely that $A\mathbf{v}_1, \dots, A\mathbf{v}_n$ are mutually orthogonal — and the σ'_i are then *forced* to be the norms $\|A\mathbf{v}_i\|$, not chosen. The freedom in Σ' is illusory; the orthogonality of the images is the entire constraint, and it is the same constraint $f(\theta) = 0$ from §2.2.

In the plane this is exactly what Figure 1 shows: of the continuum of frames, two crossings, describing one. In $\mathbb{R}^n$ the counting tells the same story. V has $n(n-1)/2$ free parameters; requiring the n images to be pairwise orthogonal imposes one equation per pair, also $n(n-1)/2$. Constraints exactly consume parameters, so the solution set is generically discrete — finitely many sign and permutation variants of one frame, not a continuum. Equivalently, $A^\top A = V\Sigma'^2V^\top$ forces V to be an eigenbasis of the symmetric matrix $A^\top A$, and eigenbases are rigid.

The exception. The degenerate case of §2.2 reappears here: when $A^\top A = \sigma^2 I$, every direction stretches equally, every orthonormal V works, and $U = AV/\sigma$. The image of the sphere is a sphere, so no direction is distinguished. The SVD tells you nothing you did not already know, and is maximally non-unique.

This is worth stating as a principle: *the content of the SVD is anisotropy*. The decomposition earns its keep exactly when the singular values differ. The ratio σ_1/σ_n is the eccentricity of the image ellipsoid, and also the condition number.

7 Uniqueness

The singular values are unique (with the descending convention $\sigma_1 \geq \dots \geq \sigma_r > 0$). The frames are not, and the freedoms are:

1. **Sign/phase.** For a simple singular value, $\mathbf{u}_i \rightarrow -\mathbf{u}_i$ and $\mathbf{v}_i \rightarrow -\mathbf{v}_i$ together. The coupling is essential: you cannot flip one without the other, since $A = U\Sigma V^\top$ must be preserved.
2. **Repeated singular values.** If σ_i has multiplicity k , the corresponding k -dimensional subspaces of U and V can be rotated by any $k \times k$ orthogonal Q — applied to *both*.
3. **Zero singular values.** The columns of U spanning $\text{coker}(A)$ and of V spanning $\ker(A)$ are arbitrary orthonormal bases of those subspaces, independent of each other.
4. **Ordering,** if the descending convention is dropped.

Claim. For A of full rank with distinct singular values, the thin SVD is unique up to the signs in item 1.

Parameter count

A general $n \times n$ matrix has n^2 free parameters. The SVD supplies

$$\underbrace{\frac{n(n-1)}{2}}_V + \underbrace{n}_\Sigma + \underbrace{\frac{n(n-1)}{2}}_U = n^2.$$

It is a reparametrization, not a compression. The information is rearranged from “what each column does” into “(input frame, stretches, output frame)”, and that arrangement is where the geometry lives.

The $n(n-1)/2$ is the dimension of the orthogonal group $O(n)$. Two ways to see it. First, $V^T V = I$ is a symmetric matrix equation, imposing $n(n+1)/2$ constraints on n^2 entries, leaving $n^2 - n(n+1)/2 = n(n-1)/2$. Second, build V column by column: $\mathbf{v}_1$ is any unit vector ($n-1$ parameters), $\mathbf{v}_2$ any unit vector orthogonal to it ($n-2$), and so on, giving $(n-1) + (n-2) + \dots + 0 = n(n-1)/2$. For $n=2$ this is 1 — a rotation is an angle. For $n=3$ it is 3 — the Euler angles.

U and V are independent

Pick any orthogonal U , any orthogonal V , any nonnegative diagonal Σ : then $A = U\Sigma V^T$ is a matrix with exactly that SVD. Nothing couples the two frames, and the parameter count above confirms there is room for both.

Going *backwards* from a given A , however, the frames are determined and generally unrelated — you cannot infer one from the other without passing through A . This is precisely why the SVD carries more information than a single eigenbasis: two independent frames rather than one.

8 Relation to eigenvalues and eigenvectors

For a general square matrix, the two decompositions are largely unrelated.

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} : \quad \lambda = 0, 0 \quad \text{but} \quad \sigma = 1, 0.$$

$$A = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : \quad \lambda = 1, 1 \quad \text{for all } t, \quad \text{but} \quad \sigma_1 \rightarrow \infty \quad \text{as } t \rightarrow \infty.$$

The structural difference: singular vectors are always orthonormal and always exist in full; eigenvectors need not be orthogonal, and a defective matrix does not have a complete set. Eigenvectors live in one space and map to themselves; singular vectors come in *coupled pairs across two spaces*. That is the fundamental distinction — eigendecomposition uses one basis, SVD pairs two.

What does hold.

- $\prod |\lambda_i| = \prod \sigma_i = |\det A|$.

- $\sigma_{\min} \leq |\lambda_i| \leq \sigma_{\max}$ for every eigenvalue; in particular the spectral radius satisfies $\rho(A) \leq \sigma_{\max} = \|A\|_2$.
- Weyl: $\prod_{i=1}^k |\lambda_i| \leq \prod_{i=1}^k \sigma_i$ for each k , with equality at $k = n$.
- Yamamoto: $\lim_{k \rightarrow \infty} \sigma_i(A^k)^{1/k} = |\lambda_i|$.

When they coincide.

- **Normal** A (i.e. $A^T A = A A^T$): $\sigma_i = |\lambda_i|$, and the eigenbasis is orthonormal, so $\mathbf{v}_i$ are the eigenvectors and $\mathbf{u}_i = \pm \mathbf{v}_i$.
- **Symmetric positive semidefinite**: $\sigma_i = \lambda_i$ and $U = V$. The two decompositions are literally identical.
- **Symmetric indefinite**: $\sigma_i = |\lambda_i|$, and U and V differ by sign flips on the negative eigendirections.

So $U = V$ characterizes symmetric positive semidefinite matrices. Neither example in this note is symmetric, which is why neither has aligned frames.

A caveat on the normal case: eigenvalues ± 3 in a symmetric matrix give a *repeated* singular value $\sigma = 3$, so the SVD acquires rotation freedom in that subspace that the eigendecomposition does not have. The eigendecomposition can be strictly more rigid.

9 Polar decomposition

For invertible square A , the SVD says

$$GL(n) = O(n) \cdot (\text{positive diagonal}) \cdot O(n),$$

which regrouped gives the polar decomposition $A = QP$ with $Q = UV^T$ orthogonal and $P = V\Sigma V^T$ positive definite. The same content in different clothes: P is the pure stretch, Q the pure rotation. For the running example Q is the rotation by 23.20° — the frame gap visible in Figure 2 — and for the shear it is 26.57° .

10 Summary

A is completely described by which orthonormal input frame goes to which orthonormal output frame, and by how much each direction stretches. The SVD's claim is that such frames always exist.

Linearity means any basis determines A — that is not what makes the SVD special. What makes the V -basis special is that in it the action is *diagonal*: feed in $\mathbf{v}_3$, get out pure $\mathbf{u}_3$, scaled. Feed in any other basis and the images tangle together in a way that hides the geometry. Same information, worse presentation.

All numerical values in this document were computed with NumPy's `linalg.svd` and verified against the defining properties ($\langle A\mathbf{v}_1, A\mathbf{v}_2 \rangle = 0$, $\prod \sigma_i = |\det A|$, $\det(UV^T) = +1$).